

Linux Practical Examination

Q1. Basic File Operations

- Create a directory named LinuxExam.
- Navigate into the directory.
- Create:
 - o student.txt
 - o course.txt
 - o result.txt
- Display the current working directory.
- Display the contents of the directory.

```
ubuntu@ip-172-31-6-190:~$ mkdir LinuxExam
ubuntu@ip-172-31-6-190:~$ cd LinuxExam
ubuntu@ip-172-31-6-190:~/LinuxExam$ touch student.txt course.txt result.txt
ubuntu@ip-172-31-6-190:~/LinuxExam$ pwd
/home/ubuntu/LinuxExam
ubuntu@ip-172-31-6-190:~/LinuxExam$ ls -l
total 0
-rw-rw-r-- 1 ubuntu ubuntu 0 Sep 26 03:39 course.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Sep 26 03:39 result.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Sep 26 03:39 student.txt
ubuntu@ip-172-31-6-190:~/LinuxExam$
```

Q2. File Management

Inside LinuxExam:

- Copy student.txt as student_backup.txt.
- Rename course.txt to linux_course.txt.
- Delete result.txt.
- Create directories:
 - o Documents
 - o Backups
 - o Scripts
- Move student_backup.txt into Backups.
- Display the directory structure.

```
ubuntu@ip-172-31-6-190:~/LinuxExam$ cp student.txt student_backup.txt
ubuntu@ip-172-31-6-190:~/LinuxExam$ mv course.txt linux_course.txt
ubuntu@ip-172-31-6-190:~/LinuxExam$ rm result.txt
ubuntu@ip-172-31-6-190:~/LinuxExam$ mkdir Documents Backups Scripts
ubuntu@ip-172-31-6-190:~/LinuxExam$ mv student_backup.txt Backups/
ubuntu@ip-172-31-6-190:~/LinuxExam$ ls -R
.:
Backups Documents Scripts linux_course.txt student.txt

./Backups:
student_backup.txt

./Documents:

./Scripts:
```

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Q3. File Content Operations

- Add at least 5 lines of student information to student.txt.
- Add at least 5 lines of course information to linux_course.txt.
- Display the contents using cat. Display the first 3 lines using head.
- Display the last 2 lines using tail.
- Count the number of lines, words, and characters in student.txt.

```
[ubuntu@ip-172-31-6-190:~/LinuxExam$ cat student.txt
Name: YOGESHWAR
Roll Number: 101
Course: Linux
College: ABC College
Semester: 3
[ubuntu@ip-172-31-6-190:~/LinuxExam$ cat linux_course.txt
Course Name: Linux Administration
Module 1: File Management
Module 2: User Management
Module 3: File Permissions
Module 4: Networking
[ubuntu@ip-172-31-6-190:~/LinuxExam$ head -n 3 student.txt
Name: YOGESHWAR
Roll Number: 101
Course: Linux
[ubuntu@ip-172-31-6-190:~/LinuxExam$ tail -n 2 student.txt
College: ABC College
Semester: 3
[ubuntu@ip-172-31-6-190:~/LinuxExam$ wc -l student.txt
5 student.txt
[ubuntu@ip-172-31-6-190:~/LinuxExam$ wc -w student.txt
12 student.txt
[ubuntu@ip-172-31-6-190:~/LinuxExam$ wc -m student.txt
80 student.txt
[ubuntu@ip-172-31-6-190:~/LinuxExam$ █
```

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Q4. User Management

- Create a user named student01.
- Set a password.
- Verify that the user exists.
- Display the user's UID and GID.
- Display the user's home directory.
- Switch to the student01 account and verify the current username.

```
[ubuntu@ip-172-31-6-190:~]$ getent passwd student01
student01:x:1002:1002::/home/student01:/bin/bash
[ubuntu@ip-172-31-6-190:~]$ id student01
uid=1002(student01) gid=1002(student01) groups=1002(student01)
[ubuntu@ip-172-31-6-190:~]$ getent passwd student01 | cut -d: -f6
/home/student01
[ubuntu@ip-172-31-6-190:~]$ sudo su - student01
[student01@ip-172-31-6-190:~]$ whoami
student01
[student01@ip-172-31-6-190:~]$ exit
logout
ubuntu@ip-172-31-6-190:~$
```

Q5. Group Management

- Create a group named linuxbatch.
- Add student01 to linuxbatch.
- Verify the groups of student01.
- Create another user named student02.
- Add student02 to linuxbatch.
- Display all members of the group.

```
[ubuntu@ip-172-31-6-190:~]$ sudo groupadd linuxbatch
[ubuntu@ip-172-31-6-190:~]$ sudo usermod -aG linuxbatch student01
[ubuntu@ip-172-31-6-190:~]$ groups student01
student01 : student01 linuxbatch
[ubuntu@ip-172-31-6-190:~]$ sudo useradd -m -s /bin/bash student02
[ubuntu@ip-172-31-6-190:~]$ sudo passwd student02
New password:
Retype new password:
passwd: password updated successfully
[ubuntu@ip-172-31-6-190:~]$ sudo usermod -aG linuxbatch student02
[ubuntu@ip-172-31-6-190:~]$ getent group linuxbatch
linuxbatch:x:1003:student01,student02
```

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Q6. File Permissions

- Create project.txt and:
- Give the owner Read, Write and Execute permission.
- Give Group Read and Execute permission.
- Give Others Read permission only.
- Verify the permissions.

Expected: 754

- Then perform the following:
 - Change the permission to 640.
 - Change the owner of the file to student01.
 - Change the group ownership to linuxbatch.
 - Verify the final ownership and permissions.

```
[ubuntu@ip-172-31-6-190:~/LinuxExam$ touch project.txt
[ubuntu@ip-172-31-6-190:~/LinuxExam$ chmod 754 project.txt
[ubuntu@ip-172-31-6-190:~/LinuxExam$ ls -l project.txt
-rwxr-xr-- 1 ubuntu ubuntu 0 Sep 26 04:35 project.txt
[ubuntu@ip-172-31-6-190:~/LinuxExam$ chmod 640 project.txt
[ubuntu@ip-172-31-6-190:~/LinuxExam$ ls -l project.txt
-rw-r----- 1 ubuntu ubuntu 0 Sep 26 04:35 project.txt
[ubuntu@ip-172-31-6-190:~/LinuxExam$ sudo chown student01:linuxbatch project.txt
[ubuntu@ip-172-31-6-190:~/LinuxExam$ ls -l project.txt
-rw-r----- 1 student01 linuxbatch 0 Sep 26 04:35 project.txt
[ubuntu@ip-172-31-6-190:~/LinuxExam$
```

Q7. Permission Challenge

- Create:
 - LinuxExam/
 - ├── public/ └── private/
 - └── shared/
- Configure permissions so that:
 - public can be accessed by everyone.
 - private can be accessed only by its owner.
 - shared can be accessed by the owner and group members.
- Verify the permissions using ls -ld.

```
[ubuntu@ip-172-31-6-190:~/LinuxExam$ cd
[ubuntu@ip-172-31-6-190:~$ mkdir public private shared
[ubuntu@ip-172-31-6-190:~$ chmod 755 public
[ubuntu@ip-172-31-6-190:~$ chmod 700 private
[ubuntu@ip-172-31-6-190:~$ chmod 770 shared
[ubuntu@ip-172-31-6-190:~$ sudo chgrp linuxbatch shared
[ubuntu@ip-172-31-6-190:~$ ls -ld public private shared
drwx----- 2 ubuntu ubuntu      4096 Sep 26 04:45 private
drwxr-xr-x 2 ubuntu ubuntu      4096 Sep 26 04:45 public
drwxrwx--- 2 ubuntu linuxbatch 4096 Sep 26 04:45 shared
[ubuntu@ip-172-31-6-190:~$
```

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Q8. Package Management

- Update the package repository.
- Install Apache Web Server.
- Verify that Apache is installed.
- Display the installed Apache version.
- Start the Apache service.
- Check the status of Apache.
- Configure Apache to start automatically at boot.

```
ubuntu@ip-172-31-6-190:~$ sudo apache2 -v
Server version: Apache/2.4.66 (Ubuntu)
Server built:   2026-07-06T15:33:20
ubuntu@ip-172-31-6-190:~$ sudo systemctl start apache2
ubuntu@ip-172-31-6-190:~$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-09-28 16:57:05 UTC; 3min 32s ago
     Invocation: c7a582184e93427b8b7e3395fbb144c4
       Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 2780 (apache2)
      Status: "Total requests: 1; Idle/Busy workers 100/0;Requests/sec: 0.00478; Bytes served/sec: 1 B/sec"
        Tasks: 55 (limit: 627)
       Memory: 8.4M (peak: 8.5M)
          CPU: 80ms
    CGroup: /system.slice/apache2.service
            └─2780 /usr/sbin/apache2 -k start -DFOREGROUND
              └─2783 /usr/sbin/apache2 -k start -DFOREGROUND
                └─2784 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 28 16:57:05 ip-172-31-6-190 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 28 16:57:05 ip-172-31-6-190 systemd[1]: Started apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-6-190:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
ubuntu@ip-172-31-6-190:~$ systemctl is-enabled apache2
enabled
ubuntu@ip-172-31-6-190:~$
```

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Q9. Apache Web Server Configuration

- Find the Apache document root.
- Create a custom index.html.
- Put the following information on the webpage:
 - o Student Name
 - o Roll Number
 - o Course Name
 - o "Linux Practical Examination"
- Start/restart Apache.
- Access the webpage using the browser or curl.
- Display the HTTP response using curl.

```
ubuntu@ip-172-31-6-190:/var/www/html$ curl -i http://localhost
HTTP/1.1 200 OK
Date: Mon, 28 Sep 2026 17:51:02 GMT
Server: Apache/2.4.66 (Ubuntu)
Last-Modified: Mon, 28 Sep 2026 17:49:07 GMT
ETag: "1f5-65c8eaebe0a1a"
Accept-Ranges: bytes
Content-Length: 501
Vary: Accept-Encoding
Content-Type: text/html

<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Linux Practical Examination</title>
</head>

<body>

  <h1>Linux Practical Examination</h1>

  <h2>Student Information</h2>

  <p><strong>Student Name:</strong> Yogeshwar Mojad</p>
  <p><strong>Roll Number:</strong> 107</p>
  <p><strong>Course Name:</strong> Linux Administration</p>

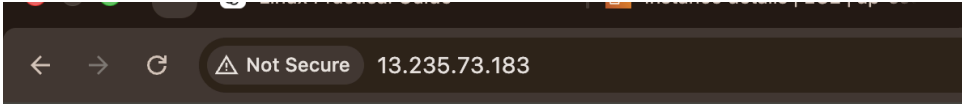
  <hr>

  <h2>Linux Practical Examination</h2>

  <p>Apache Web Server is successfully configured and running.</p>

</body>
</html>
```

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Student Information

Student Name: Yogeshwar Mojad

Roll Number: 107

Course Name: Linux Administration

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Apache Web Server is successfully configured and running.

Q10. Process Management

- Perform the following:
- Display all running processes.
- Display processes in real time using top.
- Find the PID of Apache.
- Display the process using its PID. Stop Apache.
- Verify that Apache has stopped.
- Start Apache again.
- Verify its status.
- Students should:
 - Identify the PID.
 - Display process details.
 - Terminate the process.
 - Verify that it has stopped.

```
ubuntu@ip-172-31-6-190:~$ pgrep -a apache2
3961 /usr/sbin/apache2 -k start -DFOREGROUND
3965 /usr/sbin/apache2 -k start -DFOREGROUND
3966 /usr/sbin/apache2 -k start -DFOREGROUND
ubuntu@ip-172-31-6-190:~$ ps -fp $(pgrep -o apache2)
UID      PID     PPID  C  STIME TTY          TIME CMD
root      3961      1   0  17:49 ?        00:00:00 /usr/sbin/apache2 -k start -DFOREGROUND
ubuntu@ip-172-31-6-190:~$ sudo systemctl stop apache2
ubuntu@ip-172-31-6-190:~$ systemctl is-active apache2
inactive
ubuntu@ip-172-31-6-190:~$ pgrep -a apache2
ubuntu@ip-172-31-6-190:~$ sudo systemctl start apache2
ubuntu@ip-172-31-6-190:~$ systemctl is-active apache2
active
ubuntu@ip-172-31-6-190:~$
```

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Q11. Search and Text Processing

- Inside LinuxExam:
- Create a file named students.txt containing at least 10 student records.
- Use grep to find students matching a specified name or pattern.
- Use grep with case-insensitive searching.
- Count matching records.
- Sort the student records.
- Search for a particular file using find.
- Find all students whose name contains the letter a.

Or:

- Find all .txt files inside the LinuxExam directory.

```
ubuntu@ip-172-31-6-190:~/LinuxExam$ cat > students.txt <<'EOF'
101,Amit
102,Rahul
103,Priya
104,Sneha
105,Anjali
106,Rohan
107,Akash
108,Neha
109,Sameer
110,Karan
EOF
ubuntu@ip-172-31-6-190:~/LinuxExam$ grep "Rahul" students.txt
102,Rahul
ubuntu@ip-172-31-6-190:~/LinuxExam$ grep -i "rahul" students.txt
102,Rahul
ubuntu@ip-172-31-6-190:~/LinuxExam$ grep -ic "rahul" students.txt
1
ubuntu@ip-172-31-6-190:~/LinuxExam$ sort students.txt
101,Amit
102,Rahul
103,Priya
104,Sneha
105,Anjali
106,Rohan
107,Akash
108,Neha
109,Sameer
110,Karan
ubuntu@ip-172-31-6-190:~/LinuxExam$ find . -name "student.txt"
./student.txt
```

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```
[ubuntu@ip-172-31-6-190:~/LinuxExam$ grep -i "a" students.txt
101,Amit
102,Rahul
103,Priya
104,Sneha
105,Anjali
106,Rohan
107,Akash
108,Neha
109,Sameer
110,Karan
[ubuntu@ip-172-31-6-190:~/LinuxExam$ find . -type f -name "*.txt"
./student.txt
./Backups/student_backup.txt
./linux_course.txt
./students.txt
./project.txt
ubuntu@ip-172-31-6-190:~/LinuxExam$
```

Q12. Linux Networking

- Perform the following:

- Display the system hostname.
- Display the IP address.
- Display network interfaces.
- Display the routing table.
- Test connectivity to another system or a specified website.
- Use ping to test connectivity.
- Use curl to access a webpage. Find the IP address of a domain name.
- Display listening ports.

```
[ubuntu@ip-172-31-6-190:~]$ hostname
ip-172-31-6-190
[ubuntu@ip-172-31-6-190:~]$ hostname -I
172.31.6.190
[ubuntu@ip-172-31-6-190:~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 0a:ab:6d:0f:79:a7 brd ff:ff:ff:ff:ff:ff
    altname enx0aab6d0f79a7
    inet 172.31.6.190/20 metric 100 brd 172.31.15.255 scope global dynamic ens5
        valid_lft 3332sec preferred_lft 3332sec
    inet6 fe80::8ab:6dff:fe0f:79a7/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[ubuntu@ip-172-31-6-190:~]$ ip route
default via 172.31.0.1 dev ens5 proto dhcp src 172.31.6.190 metric 100
172.31.0.0/20 dev ens5 proto kernel scope link src 172.31.6.190 metric 100
172.31.0.1 dev ens5 proto dhcp scope link src 172.31.6.190 metric 100
172.31.0.2 dev ens5 proto dhcp scope link src 172.31.6.190 metric 100
[ubuntu@ip-172-31-6-190:~]$ ping -c 4 google.com
PING google.com (172.217.160.110) 56(84) bytes of data:
64 bytes from tsao3s06-in-f14.1e100.net (172.217.160.110): icmp_seq=1 ttl=117 time=1.80 ms
64 bytes from tsao3s06-in-f14.1e100.net (172.217.160.110): icmp_seq=2 ttl=117 time=1.82 ms
64 bytes from tsao3s06-in-f14.1e100.net (172.217.160.110): icmp_seq=3 ttl=117 time=1.83 ms
64 bytes from tsao3s06-in-f14.1e100.net (172.217.160.110): icmp_seq=4 ttl=117 time=1.84 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.804/1.822/1.838/0.013 ms
```

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```
ubuntu@ip-172-31-6-190:~$ curl -I https://www.google.com
HTTP/2 200
content-type: text/html; charset=ISO-8859-1
content-security-policy-report-only: object-src 'none';base-uri 'self';script-src 'nonce-Aqq2TJwC1WaGIYH5Q0eqRQ' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http;report-uri https://csp.withgoogle.com/csp/gws/other-hp
accept-ch: Sec-CH-Prefers-Color-Scheme
p3p: CP="This is not a P3P policy! See g.co/p3phelp for more info."
date: Mon, 28 Sep 2026 18:07:05 GMT
server: gws
x-xss-protection: 0
x-frame-options: SAMEORIGIN
expires: Mon, 28 Sep 2026 18:07:05 GMT
cache-control: private
set-cookie: __Secure-STRP=ANmZwa18ewu3i0a27xqjakZhU_33Q7LYfPxLkmwk7QU3Twder3X9oGXUgFXJk6PxaSyP8CbFAsCJqpWy3UNWwA_Y69PpCrs41Hw; expires=Mon, 28-Sep-2026 18:12:05 GMT; path=/; domain=.google.com; Secure; SameSite=strict
set-cookie: AEC=Aaa9EJoHXNE60v_vQ18dQ2pqi3Xoyx09MGc-_VQzBi90ddvvnVBxZ8estz0; expires=Sat, 27-Mar-2027 18:07:05 GMT; path=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
set-cookie: NID=Cu8BCAESoQE80xGDSNclle4d8h_KUhJqNVRMvQliGep6e7X1PLCsABmaxy2W4BE0vD800A_MLVLFJTDBu1Zvum4MzZgk31L9hFZBG4hy29SDPYxnt6EqT84iRz5gSRrQ66YSuw3Y1ChTQgrobSSSt2615KVcb8xeLcl0-xLMAIEzury7H-qSjaZQywpZws8r_Fc1MQAcbMIWSBfPD5rxFaHVXMJukcvh3igBMkUB0qwHz7LC7guFImXL65DZs5Q2TI50DAzu2aIW_8pkhhzdPZUX7Mb3KIIschq3oeewB5acXPU5XVdodKx2oVhcTokxCfQk; expires=Tue, 30-Mar-2027 18:07:05 GMT; path=/; domain=.google.com; HttpOnly
alt-svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

ubuntu@ip-172-31-6-190:~$ getent ahostsv4 google.com
172.217.160.110 STREAM google.com
172.217.160.110 DGRAM
172.217.160.110 RAW
ubuntu@ip-172-31-6-190:~$ sudo ss -tulnp
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port
Process
udp        UNCONN     0           0           127.0.0.1:323           0.0.0.0:*
  users:((("chronyd",pid=821,fd=4))
udp        UNCONN     0           0           127.0.0.54:53           0.0.0.0:*
  users:((("systemd-resolve",pid=186,fd=18))
udp        UNCONN     0           0           127.0.0.53%lo:53        0.0.0.0:*
  users:((("systemd-resolve",pid=186,fd=16))
udp        UNCONN     0           0           172.31.6.190%ens5:68     0.0.0.0:*
  users:((("systemd-network",pid=595,fd=32))
udp        UNCONN     0           0           [::]:323                [::]:*
  users:((("chronyd",pid=821,fd=5))
tcp        LISTEN     0           4096        0.0.0.0:22               0.0.0.0:*
  users:((("sshd",pid=735,fd=3),("systemd",pid=1,fd=270))

tcp        LISTEN     0           4096        127.0.0.54:53           0.0.0.0:*
  users:((("systemd-resolve",pid=186,fd=19))
tcp        LISTEN     0           4096        127.0.0.53%lo:53        0.0.0.0:*
  users:((("systemd-resolve",pid=186,fd=17))
tcp        LISTEN     0           80         127.0.0.1:3306           0.0.0.0:*
  users:((("mariadb",pid=943,fd=29))
tcp        LISTEN     0           511         *:80                     *:~
  users:((("apache2",pid=4172,fd=4),("apache2",pid=4171,fd=4),("apache2",pid=4168,fd=4))
tcp        LISTEN     0           4096        [::]:22                  [::]:*
  users:((("sshd",pid=735,fd=4),("systemd",pid=1,fd=271))
ubuntu@ip-172-31-6-190:~$ systemctl is-active apache2
active
ubuntu@ip-172-31-6-190:~$ sudo ss -tulnp | grep ':80 '
tcp        LISTEN     0           511         *:80                     *:~      users:((("apache2",pid=4172,fd=4),("apache2",pid=4171,fd=4),("apach
e2",pid=4168,fd=4))
ubuntu@ip-172-31-6-190:~$ curl -I http://localhost
HTTP/1.1 200 OK
Date: Mon, 28 Sep 2026 18:09:09 GMT
Server: Apache/2.4.66 (Ubuntu)
Last-Modified: Mon, 28 Sep 2026 17:49:07 GMT
ETag: "1f5-65c8eaebe0a1a"
Accept-Ranges: bytes
Content-Length: 501
Vary: Accept-Encoding
Content-Type: text/html
```